

Europass Curriculum Vitae



Personal information

Name / Surname

Address

Personal Email

Home page

Nationality

Gender

Domenico / Di Sante

Department of Physics and Astronomy, Alma Mater Studiorum – University of Bologna, Viale C. Berti Pichat 6/2, 40127 Bologna, Italy
domenico.disante@unibo.it

<https://www.unibo.it/sitoweb/domenico.disante/en>

Italian

Male

Employment

Dates

Position

Institute

1 February 2024 - present

Associate Professor

Department of Physics and Astronomy, University of Bologna

Dates

Position

Institute

1 February 2021 - 31 January 2024

Assistant Professor (tenure track)

Department of Physics and Astronomy, University of Bologna

Dates

Position

Institute

1 December 2016 - 6 December 2020

Akademischer Rat (Academic Scientist)

Institut für Theoretische Physik und Astrophysik, University of Würzburg – Supervision Prof. Ronny Thomale

Dates

Position

Institute

17 February 2016 - 30 November 2016

PostDOC

Institut für Theoretische Physik und Astrophysik, University of Würzburg – Supervision Prof. Giorgio Sangiovanni

Dates

Position

Institute

17 November 2014 - 16 February 2016

PostDOC

CNR-SPIN – Supervision Prof. Silvia Picozzi

Fellowships and Affiliations

2024-2027	Guest Researcher at the Max Planck Institute for the Structure and Dynamics of Mat- ters, Hamburg
2023-2027	Mercator Fellow of the Collaborative Research Center SFB 1170 "ToCoTronics" (Project-ID 258499086) of the Deutsche Forschungsgemeinschaft (DFG)
2023-2025	Associate Researcher at CNR-IOM Trieste
2021-2022	EU Marie-Curie Global Fellow at the Center for Computational Quantum Physics, Flatiron Institute, New York – Supervision Prof. Andrew Millis

Education

Date	May 2021
Degree	Habilitation as Italian Professor
Category	Full Professor in Theoretical Condensed Matter Physics (02/B2 FIS/03)
Date	April 2017
Degree	Habilitation as Italian Professor
Category	Associate Professor in Theoretical Condensed Matter Physics (02/B2 FIS/03)
Date	02 November 2011 - 31 October 2014
Degree	PhD degree in Physics (15 April 2015)
Thesis	Modeling Cross Coupling Interactions in Advanced Materials. Spin-Orbit, Multiferroic- ity, Disorder and Electron-Phonon Interaction.
Supervisors	Dr. Silvia Picozzi and Prof. Sergio Ciuchi
University	Department of Physical and Chemical Sciences, University of L'Aquila
Date	10 October - 09 November 2011
	Research Fellowship with title "Studio teorico-computazionale dell'accoppiamento magneto-elettrico all'interfaccia Fe/BaTiO ₃ con funzionali avanzati di scambio e corre- lazione"
	CNR-SPIN (Dr. S. Picozzi)
Date	01 November 2009 - 21 July 2011
Degree	Master Degree in Physics
University	University of L'Aquila
Final Mark	110/110 cum laude
Date	October 2006 - October 2009
Degree	Bachelor Degree in Physics
University	University of L'Aquila
Final Mark	110/110
Date	September 2001 - July 2006
Degree	High-School
School	Liceo Scientifico A. Einstein (Teramo)
Final Mark	100/100

Third-party funding

MUR FIS-2023-00144 (2026-2029)	Frontiers of kagome metals and line-graph lattices – FrontLine, CUP J53C25001880001. Budget: 1.651.270,90 € .
DFG-SFB1170 (2023-2027)	Mercator Fellowship. Budget: ~ 38 k€ .
EU project 897276 (2021-2024)	Marie Curie Global Fellowship Horizon 2020-MSCA-IF-2019, Principal Investigator of the project BITMAP "aB-IniTio calculations and MACHine learning for suPERconducting collective phenomena in novel materials". Budget: ~ 270 k€ .
DFG-SFB1170 (2019-2023)	Principal Investigator of the project "Topological Fermi surface instabilities from a combined ab initio and functional renormalization group workflow". Budget: ~ 398 k€ .

Prizes and Awards

2019	Nomination in the Emerging Leaders 2020 by the JPhys Materials (IOP Publishing).
2006-2009	Student Scholarship of 12 k€ from the Italian Physics Society (SIF).
2005	Prize of the National Laboratories of GranSasso for a summer school at the Princeton University.

Research interests

Machine Learning for correlated electrons systems
Hydrodynamics of Dirac electron fluids
Topological states in 3 and 2 dimensional systems
Topological superconductivity and unconventional Fermi surface instabilities
Correlated higher-order Dirac semimetals
Ab initio calculations of Ferroelectrics and Multiferroics
Ferroelectric Rashba Semiconductors
Electron-phonon coupling in disordered systems

Institutional Responsibilities

Vice-coordinator	Corso di Studi triennale in Scienza dei Materiali
Member of Commission	Admission to the PhD program in Physics (40 cycle)
Erasmus+	Responsible for the University of Bologna–University TU-Wien exchange program, 2025-2029
Erasmus+	Responsible for the University of Bologna–University of Würzburg exchange program, 2024-2028
Member of Commission	Joint Committee (Commissione Paritetica) for the assessment of the formative offer and the quality of teaching, 2024-2027
Member of Commission	Master's Degree in Physics, March 26 th 2024, University of Bologna
Member of Commission	Final exam of PhD Degree in Physics, March 21 st 2024, University of Bologna
Member of Commission	PostDoc selection, January 9 th 2024, University of Bologna

Member of Commission	Master's Degree in Physics, October 19 th 2023, University of Bologna
RFO-2023	Coordinator of funds (30k€) for the Matter sector
President of Commission	Final exam of PhD Degree in Physics, June 15 th 2023, University of Bologna
Member of Commission	Master's Degree in Physics, March 30 th 2023, University of Bologna
President of Commission	Final exam of PhD Degree in Physics, March 15 th 2023, University of Bologna
Member of Commission	PostDoc selection, January 18 th 2023, University of Bologna
Third Mission	
PLS	Responsabile Unibo del Piano Lauree Scientifiche in Scienza dei Materiali
2023	Realization of the promotional video for the bachelor degree in Materials Science. Link to the video
Teaching	
Date	Academic Year 2025/2026, Summer semester
Course	Fisica della Materia (52 hours)
Where	University of Bologna, Bachelor Degree in Physics
Date	Academic Year 2025/2026, Summer semester
Course	Interactions and Correlations in Condensed Matter (32 hours)
Where	University of Bologna, Master Degree in Physics
Date	Academic Year 2025/2026, Winter semester
Course	Data Science (56 hours)
Where	University of Bologna, Bachelor Degree in Scienza dei Materiali
Date	Academic Year 2024/2025, Summer semester
Course	Fisica della Materia (52 hours)
Where	University of Bologna, Bachelor Degree in Physics
Date	Academic Year 2024/2025, Summer semester
Course	Interactions and Correlations in Condensed Matter (32 hours)
Where	University of Bologna, Master Degree in Physics
Date	Academic Year 2024/2025, Winter semester
Course	Data Science (64 hours)
Where	University of Bologna, Bachelor Degree in Scienza dei Materiali
Date	Academic Year 2023/2024, Summer semester
Course	Interactions and Correlations in Condensed Matter (32 hours)
Where	University of Bologna, Master Degree in Physics
Date	Academic Year 2023/2024, Winter semester
Course	Computational Material Physics (28 hours)
Where	University of Bologna, Master Degree in Physics
Date	Academic Year 2022/2023, Summer semester

Course	Interactions and Correlations in Condensed Matter (32 hours)
Where	University of Bologna, Master Degree in Physics
Date	Academic Year 2019/2020, Summer semester
Course	Theory of Superconductivity (54 hours)
Where	Wuerzburg University, Master Degree in Physics
Date	Academic Year 2019/2020, Winter semester
Course	Computational Material Science (34 hours)
Where	Wuerzburg University, Master Degree in Physics
Date	Academic Year 2018/2019, Summer semester
Course	Classical Electrodynamics, Exercises (24 hours)
Where	Wuerzburg University, Bachelor Degree in Physics
Date	Academic Year 2018/2019, Winter semester
Course	Computational Material Science (34 hours)
Where	Wuerzburg University, Master Degree in Physics
Date	Academic Year 2017/2018, Summer semester
Course	Classical Electrodynamics, Exercises (24 hours)
Where	Wuerzburg University, Bachelor Degree in Physics
Date	Academic Year 2017/2018, Winter semester
Course	Computational Material Science (32 hours)
Where	Wuerzburg University, Master Degree in Physics
Date	Academic Year 2016/2017, Summer semester
Course	Quantum Mechanics II, Exercises (22 hours)
Where	Wuerzburg University, Bachelor Degree in Physics
Date	Academic Year 2016/2017, Winter semester
Course	Computational Material Science (26 hours)
Where	Wuerzburg University, Master Degree in Physics

Thesis Co/Supervision

Name	Lorenzo Riguzzi, University of Bologna (Master thesis), 2026
Topic	Study of electronic correlations in Cr based Kagome metals via slave spin mean field
Role	Supervisor
Name	Ahmed Osman, University of Bologna (Master thesis), 2026
Topic	Efficient and GPU-Accelerated Implementation of Berry-Phase-Related Operators in Electronic-Structure Calculations
Role	Supervisor
Name	Libero Pollini, University of Bologna (Bachelor thesis), 2025
Topic	Reti Neurali "Physics-Informed" e loro applicazione alle equazioni differenziali
Role	Supervisor
Name	Erio Quadrelli, University of Bologna (Bachelor thesis), 2025
Topic	Teoria della fase di Berry, conseguenze topologiche e applicazioni
Role	Supervisor

Name	Stefano Bilamour, University of Bologna (Master thesis), 2025
Topic	Geodesic dynamics of Bloch's electrons in twisted Moiré heterostructures
Role	Supervisor
Name	Chiara Pacella, University of Bologna (Master thesis), 2024
Topic	Orbital Hall effect in Weyl semimetals
Role	Supervisor
Name	Lorenzo Riguzzi, University of Bologna (Bachelor thesis), 2024
Topic	First principles calculations of electronic properties of CsV ₃ Sb ₅ kagome metal under doping
Role	Supervisor
Name	Francesco Nasoni, University of Bologna (Bachelor thesis), 2023
Topic	Calcolo a primi principi dello spettro di assorbimento di perovskiti per celle solari
Role	Supervisor
Name	Samuele De Amicis, University of Bologna (Bachelor thesis), 2023
Topic	La superconduttività dalla teoria BCS tradizionale al caso dei cuprati con annessa risoluzione numerica dell'equazione di gap in condizione di interazione anisotropa
Role	Co-supervisor
Name	Pasquale Santodirocco, University of Bologna (Bachelor thesis), 2022
Topic	Calcoli analitici ed a principi primi di tipiche strutture elettroniche per reticolo triangolare, esagonale e kagome
Role	Co-supervisor
Name	Alessandro Ciavatta, University of Bologna (Bachelor thesis), 2021
Topic	Teoria della superconduttività e soluzione numerica dell'equazione della gap BCS
Role	Co-supervisor

Bibliometric Indicators

h-index	44 (Google Scholar)
Total Citations	6645 (Google Scholar)

Invited Talks at Conferences

Date	5-7-8 July 2026
Conferene	International School of Physics "Enrico Fermi" Quantum Matter, Varenna, Italy
Title	Lectures on Kagome Metals: Geometry, Topology and Emergent Quantum Phenomena
Date	13 January 2026
Conferene	International Workshop on Spin–Orbit Driven Phenomena in Quantum Matter, Bologna, Italy
Title	Spin-Berry Curvature in Kagome Metals
Date	5 November 2025
Conferene	IOM Workshop, Trieste, Italy
Title	The rich physics of kagome metals
Date	10 July 2025
Conferene	FISMAT, Venice, Italy

Title	The rich physics of kagome metals
Date	15 June 2025
Conferene	Faculty Colloquium, Würzburg, Germany
Title	The rich physics of kagome metals
Date	13 March 2025
Conferene	Advancing the Frontier of Kagome Materials, Zurich, Switzerland
Title	The rich physics of bylayer kagome metals
Date	3 December 2024
Conferene	QUEST Conference, Trieste, Italy
Title	The rich physics of bylayer kagome metals
Date	9 September 2024
Conferene	Congresso SIF, Bologna, Italy
Title	The rich physics of bylayer kagome metals
Date	5 June 2024
Conferene	Qu-Spin Conference, Trondheim, Norway
Title	The rich physics of bylayer kagome metals
Date	18 April 2024
Conferene	MADYCS'24, Trieste, Italy
Title	The rich physics of kagome metals
Date	18 September 2023
Conferene	E-MRS Fall Meeting, Warsaw, Poland
Title	Emergent honeycomb physics from chiral atomic orbitals on a triangular lattice
Date	31 August 2023
Conferene	ECOSS36, 36th European Conference on Surface Science, Lodz, Poland
Title	The Orbital Angular Momentum to shed light onto the Topology of Quantum Materials
Date	13 July 2023
Conferene	Workshop on Complexity and Topology in Quantum Matter, Würzburg, Germany
Title	The physics of $p_x p_y$ orbitals on the triangular and hexagonal lattices. The cases of Indene and Bismuthene
Date	5 June 2023
Conferene	CMT@BRIXEN, Bressanone, Italy
Title	Deep Learning the Functional Renormalization Group
Date	16 June 2022
Conferene	Challenges and Prospects of ML for the Physical Sciences, New York, USA
Title	Deep Learning the Functional Renormalization Group
Date	13 April 2022
Conferene	Lectiones Amalfitanae, Amalfi, Italy
Title	Deep Learning the Functional Renormalization Group
Date	23 January 2020
Conferene	ECOS, Milan, Italy

Title	Robust spin-polarized midgap states at step edges of topological crystalline insulators
Date	6 September 2018
Conferene	NGSCES 2018, San Sebastian, Spain
Title	Engineering the stability of Unconventional Superconductivity on hexagonal lattices
Date	31 August 2018
Conferene	Lectiones Clitumnaliae, Campello, Italy
Title	Engineering the stability of Unconventional Superconductivity on hexagonal lattices
Date	17 March 2018
Conferene	IWEPNM 2018, Kirchberg, Austria
Title	Robust spin-polarized midgap states at step edges of topological crystalline insulators
Date	3 March 2017
Conferene	Physics at the borderline between 1D and 2D, Bad Honnef, Germany
Title	Robust spin-polarized midgap states at step edges of topological crystalline insulators
Date	23 June 2016
Where	CNNEM-2016, Shanghai, China
Title	Emergence of ferroelectricity and spin-valley properties in two-dimensional honey-comb binary compounds

Organization of Conferences

International School of Physics "Enrico Fermi" Quantum Matter	July 1 th - July 10 nd 2026, Varenna, Italy
International School of Physics "Enrico Fermi" Topology and Materials	July 16 th - July 22 nd 2024, Varenna, Italy
Kagome metals: recent breakthroughs and future perspectives	September 5 th 2023, Milan, Italy (Colloquium in the CMD30-FISMAT Conference)
2022 Flatiron Machine Learning X Science Summer School	June 6 th - August 5 th 2022, New York, USA
NGSCES 2019	September 2 nd -6 th 2019, Pescara, Italy

Editorial Commitments

Springer Proceedings in Physics	Proceedings of the International School of Physics "Enrico Fermi" – Course 215: Topology and Materials – Villa Monastero, Varenna on Lake Como, Italy, 17-22 July 2024
The European Physical Journal Plus	Editor of the Focus Point on "Machine Learning for Materials Physics: From Pitfalls to Best Practices"

Refereeing

Referee for Nature, Nature Physics, Nature Communications, Physical Review Letters, Physical Review B, Physical Review X, Journal of American Chemical Society and New Journal of Physics

Selected Publications

- 13 D. Di Sante et al., *Kagome metals*, Rev. Mod. Phys. **98**, 015002 (2026)
- 12 C. Bigi, [...], D. Di Sante *Pomeranchuk instability from electronic correlations in CsTi3Bi5 kagome metal*, Nat. Commun. **17**, 325 (2026)
- 11 H. Li, [...], D. Di Sante, Z. Wang and I. Zeljkovic *Spin Berry curvature-enhanced orbital Zeeman effect in a kagome metal*, Nat. Phys. **20**, 1103 (2024)
- 10 D. Di Sante et al., *Flat band separation and robust spin Berry curvature in bilayer kagome metals*, Nat. Phys. **19**, 1135 (2023)
- 9 D. Di Sante et al., *Deep Learning the Functional Renormalization Group*, Phys. Rev. Lett. **129**, 136402 (2022)
- 8 M. Syperek, [...], D. Di Sante and C. Schneider, *Observation of room temperature excitons in an atomically thin topological insulator*, Nat. Commun. **13**, 6313 (2022)
- 7 S. Wolf, D. Di Sante et al., *Triplet Superconductivity from Nonlocal Coulomb Repulsion in an Atomic Sn Layer Deposited onto a Si (111) Substrate*, Phys. Rev. Lett. **128**, 167002 (2022)
- 6 M. Kang, [...], D. Di Sante et al., *Twofold van Hove singularity and origin of charge order in topological kagome superconductor CsV₃Sb₅*, Nat. Phys. **18**, 301 (2022)
- 5 M. Ünzelmann, [...], D. Di Sante and F. Reinert, *Momentum-space signatures of Berry flux monopoles in a Weyl semimetal*, Nat. Commun. **12**, 3650 (2021)
- 4 D. Di Sante et al., *Turbulent hydrodynamics in strongly correlated Kagome metals*, Nat. Commun. **11**, 3997 (2020)
- 3 D. Di Sante et al., *Disorder-driven metal-insulator transitions in deformable lattices*, Phys. Rev. Lett. **118**, 036602 (2017)
- 2 P. Sessi, D. Di Sante et al., *Robust spin-polarized midgap states at step edges of topological crystalline insulators*, Science **354**, 1269 (2016)
- 1 D. Di Sante et al., *Electric Control of Giant Rashba Effect in Bulk GeTe*, Adv. Mater. **25**, 509 (2013)

Full Publications List

Articles

- 95 S. Enzner, J. Berges, A. Schobert, D. Oh, M. Kang, R. Comin, R. Thomale, T. Wehling, D. Di Sante and G. Sangiovanni *Phonon fluctuation diagnostics: Origin of charge order in AV₃Sb₅ kagome metals*, Phys. Rev. B, **114**, 144304 (2026)
- 94 X. Cai, A. Consiglio, D. Di Sante, R. Thomale, W. Hanke *Continuum theory for topological phase transitions in exciton systems*, Phys. Rev. B, **113**, 205121 (2026)

- 93 F. Mazzola, W. Brzezicki, C. Bigi, A. Consiglio, L. J. D'Onofrio, M. Teresa Mercaldo, A. Klosinski, F. Bertran, P. Le Fèvre, O. J. Clark, M. T. Edmonds, M. Tuniz, A. De Vita, V. Polewczyk, J. B. Jacobsen, H. Jacobsen, J. A. Miwa, J. W. Wells, A. Jana, I. Vobornik, J. Fujii, N. Mignani, N. T. Samani, A. Crepaldi, G. Sangiovanni, A. Kataria, T. Morresi, S. Sanna, P. Bonfá, B. R. Ortiz, G. Pokharel, S. D. Wilson, D. Di Sante, C. Ortix, M. Cuoco *Anomalous spin-optical helical effect in Ti-based kagome metal*, *Advanced Materials*, **38**, e22533 (2026)
- 92 D. Di Sante, T. Neupert, G. Sangiovanni, R. Thomale, R. Comin, J. G. Checkelsky, I. Zeljkovic, S. D. Wilson *Kagome metals*, *Rev. Mod. Phys.*, **98**, 015002 (2026)
- 91 C. Bigi, M. Dürnagel, L. Klebl, A. Consiglio, G. Pokharel, M. Zonno, F. Bertran, P. Le Fèvre, T. Jaouen, H. C. Tchoukem, P. Turban, A. De Vita, J. A. Miwa, J. W. Wells, D. Oh, R. Comin, R. Thomale, I. Zeljkovic, B. R. Ortiz, S. D. Wilson, G. Sangiovanni, F. Mazzola, D. Di Sante, *Pomeranchuk instability from electronic correlations in CsTi3Bi5 kagome metal*, *Nat. Commun.*, **17**, 325 (2026)
- 90 S. Chaudhary, A. Consiglio, J. Singh, D. Di Sante, R. Thomale, Y. Singh, *Electronic structure and phonon properties in the kagome metals RT3B2 (R=Y,Lu; T=Co,Os)*, *Phys. Rev. B*, **112**, 235170 (2025)
- 89 A. Wenger, A. Consiglio, H. Hohmann, M. Dürnagel, F. O. von Rohr, H. D. Scammell, J. Ingham, D. Di Sante, R. Thomale, *Theory of unconventional magnetism in a Cu-based kagome metal*, *Phys. Rev. B*, **112**, 214416 (2025)
- 88 M. Tuniz, A. Consiglio, G. Pokharel, F. Parmigiani, T. Neupert, R. Thomale, S. K. Chaluvadi, P. Orgiani, G. Sangiovanni, S. D. Wilson, I. Vobornik, F. Salvador, F. Cilento, D. Di Sante and F. Mazzola, *Strain-Induced Enhancement of the Charge Density Wave in the Kagome Metal ScV6Sn6*, *Phys. Rev. Lett.*, **134**, 066501 (2025)
- 87 M. Medvidovic, J. C. Umana, I. Ahmadabadi, D. Di Sante, J. Flick and A. Rubio, *Neural network distillation of orbital dependent density functional theory*, *Phys. Rev. Research*, **7**, 023113 (2025)
- 86 L. Marchetti, M. Bunney, D. Di Sante and S. Rachel, *Electronic structure, spin-orbit interaction, and electron-phonon coupling of triangular adatom lattices on semiconductor substrates*, *Phys. Rev. B*, **111**, 125115 (2025)
- 85 F. Jakubczyk, A. Consiglio, D. Di Sante, R. Thomale and C. Timm, *Orbital-selective spin-triplet superconductivity in infinite-layer LaNiO2*, *Phys. Rev. B*, **112**, L100503 (2025)
- 84 T. Figgemeier, M. Ünzelmann, P. Eck, J. Schusser, L. Crippa, J. N. Neu, B. Geldiyev, P. Kagerer, J. Buck, M. Kalläne, M. Hoesch, K. Rosnagel, T. Siegrist, L.-K. Lim, R. Moessner, G. Sangiovanni, D. Di Sante, F. Reinert and H. Bentmann, *Imaging Orbital Vortex Lines in Three-Dimensional Momentum Space*, *Phys. Rev. X*, **15**, 011032 (2025)
- 83 J. Erhardt, M. Iannetti, F. Dominguez, E. M. Hankiewicz, B. Trauzettel, G. Profeta, D. Di Sante, G. Sangiovanni, S. Moser and Ralph Claessen, *Backscattering in topological edge states despite time-reversal symmetry*, *Nat. Commun.*, **16**, 8209 (2025)
- 82 D. Di Sante and A. M. Sengupta, *Focus point on machine learning for materials physics: from pitfalls to best practice*, *Eur. Phys. J. Plus*, **140**, 778 (2025)

- 81 J. Zang, M. Medvidovic, D. Kiese, D. Di Sante, A. M. Sengupta and A. J. Millis, *Machine learning-based compression of quantum many body physics: PCA and autoencoder representation of the vertex function*, Mach. Learn.: Sci. Technol., **5**, 045076 (2024)
- 80 C. Lin, A. Consiglio, O. K. Forslund, J. Küspert, M. M. Denner, H. Lei, A. Louat, M. D. Watson, T. K. Kim, C. Cacho, D. Carbone, M. Leandersson, C. Polley, T. Balasubramanian, D. Di Sante, R. Thomale, Z. Guguchia, G. Sangiovanni, T. Neupert and J. Chang, *Uniaxial strain tuning of charge modulation and singularity in a kagome superconductor*, Nat. Commun., **15**, 10466 (2024)
- 79 C. Schmitt, S. Sotgiu, S. Enzner, J. Erhardt, E. Stellino, D. Di Sante, G. Sangiovanni, R. Claessen, S. Moser and L. Baldassarre, *All-optical quality control of indenene intercalation into graphene/SiC*, Appl. Phys. Lett. **125**, 223102 (2024)
- 78 E. Moghadas, N. Dräger, A. Toschi, J. Zang, M. Medvidovic, D. Kiese, A. J. Millis, A. M. Sengupta, S. Andergassen and D. Di Sante, *Compressing the two-particle Green's function using wavelets: Theory and application to the Hubbard atom*, Eur. Phys. J. Plus **139**, 700 (2024)
- 77 H. Li, S. Cheng, G. Pokharel, P. Eck, C. Bigi, F. Mazzola, G. Sangiovanni, S. D. Wilson, D. Di Sante, Z. Wang and I. Zeljkovic, *Spin Berry curvature-enhanced orbital Zeeman effect in a kagome metal*, Nat. Phys. **20**, 1103 (2024)
- 76 A. Consiglio, G. Gatti, E. Martino, and L. Moreschini, J. C. Johannsen, K. Prsa, P.G. Freeman, D. Sheptyakov, H. M. Ronnow, R. Scopelliti, A. Magrez, L. Forró, C. Schmitt, V. Jovic, C. Jozwiak, A. Bostwick, E. Rotenberg, T. Hofmann, R. Thomale, G. Sangiovanni, D. Di Sante, M. Greiter, M. Gioni and S. Moser, *Electron Glass Phase with Resilient Zhang-Rice Singlets in LiCu3O3*, Phys. Rev. Lett. **132**, 126502 (2024)
- 75 M. Tuniz, A. Consiglio, D. Puntel, C. Bigi, S. Enzner, G. Pokharel, P. Orgiani, W. Bronsch, F. Parmigiani, V. Polewczyk, P. D. C. King, J. W. Wells, I. Zeljkovic, P. Carrara, G. Rossi, J. Fujii, I. Vobornik, S. D. Wilson, R. Thomale, T. Wehling, G. Sangiovanni, G. Panaccione, F. Cilento, D. Di Sante and F. Mazzola, *Dynamics and resilience of the unconventional charge density wave in ScV6Sn6 bilayer kagome metal*, Commun. Mater. **4**, 103 (2023)
- 74 N. Wagner, L. Crippa, A. Amaricci, and P. Hansmann, M. Klett, E. J. König, T. Schäfer, D. Di Sante, J. Cano, A. J. Millis, A. Georges and G. Sangiovanni, *Mott insulators with boundary zeros*, Nat. Commun. **14**, 7531 (2023)
- 73 D. Di Sante, C. Bigi, P. Eck, S. Enzner, A. Consiglio, G. Pokharel, P. Carrara, P. Orgiani, V. Polewczyk, J. Fujii, P. D. C. King, I. Vobornik, G. Rossi, I. Zeljkovic, S. D. Wilson, R. Thomale, G. Sangiovanni, G. Panaccione and F. Mazzola, *Flat band separation and robust spin Berry curvature in bilayer kagome metals*, Nat. Phys. **19**, 1135 (2023)
- 72 F. Mazzola, S. Enzner, P. Eck, C. Bigi, M. Jugovac, I. Cojocariu, V. Feyer, Z. Shu, G. M. Pierantozzi, A. De Vita, P. Carrara, J. Fujii, P. D. C. King, G. Vinai, P. Orgiani, C. Cacho, M. D. Watson, G. Rossi, I. Vobornik, T. Kong, D. Di Sante, G. Sangiovanni and G. Panaccione, *Observation of Termination-Dependent Topological Connectivity in a Magnetic Weyl Kagome Lattice*, Nano Lett. **23**, 8035 (2023)
- 71 B. Geldiyev, M. Ünzelmann, P. Eck, T. Kisslinger, J. Schusser, T. Figgemeier, P. Kagerer, N. Tezak, M. Krivenkov, A. Varykhalov, A. Fedorov, L. Nicolaï, J. Minár, K. Miyamoto, T. Okuda, K. Shimada, D. Di Sante, G. Sangiovanni, L. Hammer, M. A. Schneider, H. Bentmann, and F. Reinert, *Strongly anisotropic spin and orbital Rashba effect at a tellurium-noble metal interface*, Phys. Rev. B **108**, L121107 (2023)

- 70 R. C. Vidal, G. Marini, L. Lunczer, S. Moser, L. Fürst, J. Issing, C. Jozwiak, A. Bostwick, E. Rotenberg, C. Gould, H. Buhmann, W. Beugeling, G. Sangiovanni, D. Di Sante, G. Profeta, L. W. Molenkamp, H. Bentmann, and F. Reinert, *Topological band inversion in HgTe(001): Surface and bulk signatures from photoemission*, Phys. Rev. B **107**, L121102 (2023)
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